

M6 Final Release Report

Benchmark-driven zoning bylaw verification for Burnaby R1, with Vancouver and Calgary as generalization checks.

Core contract: extraction proposes; deterministic verification proves; GIS consumes only verified, source-supported rules.

Executive Status

M6 is release-ready as a safety-first verifier and reviewer communication system. It is not a claim of complete legal-rule extraction. The system intentionally routes uncertain, incomplete, or table-ambiguous claims to review instead of promoting them.

Check	Result
Full unit suite	475 passed
Adversarial poisoned rule cases	24 / 24 blocked
Adversarial poisoned bundle cases	3 / 3 blocked
Leaks	0
Fresh M6 run	m6_final_20260616, hard gates passed

Final Safety Metrics

City	Verified rules	Effective slots	Release coverage	Precision	False verified	False approval
Burnaby R1	84	82	0.975	1.00	0	0
Vancouver RS	12	7	1.000	1.00	0	0
Calgary RCG	11	8	1.000	1.00	0	0

Too Much / Too Little Count Audit

M6 uses the scored legal denominator for reviewer-facing count trust. Raw slots remain visible only as an advisory over-count guardrail.

City	Scored legal slots	Scored verified	Coverage	Review	Missed
Burnaby R1	66	21	31.8%	3	42
Vancouver RS	44	4	9.1%	1	38
Calgary RCG	671	7	1.0%	5	628

Interpretation: M6 is not over-verifying. Unsupported verified rules are zero and duplicate verified slots are zero. The system is conservative: Burnaby table proof completion is the top remaining improvement area.

M6 Acceptance Contract

- verified_precision = 1.00
- false_verified_count = 0
- false_approval_count = 0
- verified_source_support_failed_count = 0
- verified_slot_mapping_rate = 1.00
- unsupported_verified_rule_count = 0
- duplicate_verified_slot_count = 0
- effective verified slot count does not regress from M4
- extraction_coverage_recall $\geq$ 0.95
- runtime under 300 seconds
- adversarial benchmark passes

Metric Naming

raw_candidate_artifact_recall is diagnostic only. It counts only gold rules still visible in rule_candidates.json.

release_candidate_recall / **extraction_coverage_recall** is the M6 release gate. It counts gold rules surfaced anywhere in the output path: candidate, verified, review, rejected, or not-used.

This distinction fixes the M5.6 confusion where Burnaby showed low raw candidate artifact recall but high release extraction coverage because many rules had already moved into verifier output buckets.

Reviewer Workflow

- 1. Start With Safety:** Show zero false verified rules, zero false approvals, and verified precision 1.00.
- 2. Explain Counts:** Use the Too Much / Too Little panel. Scored legal slots are the count authority; raw slots are advisory.
- 3. Show Verified Rules:** Verified rules are GIS-ready because value, unit, operator, scope, and evidence support are checked.
- 4. Show Review Queue:** Review-needed rules are not failures. They are plausible but not proven enough for GIS.
- 5. Use Chat Carefully:** Ask The Bylaw can explain source text and verified outputs. It cannot approve, reject, verify, edit, or promote anything.

Known Limits And Next Work

- Burnaby scored verified coverage is 31.8%; the project is safety-ready, not complete-rule-extraction complete.
- Burnaby table proof completion is 37 / 81, below the 50% optimization target.
- Calgary has a broad scored denominator because the full bylaw is much larger than the product scope.
- Future improvements should strengthen table evidence assembly and retrieval, never loosen verification thresholds.

Runbook

See docs/M6_FINAL_RELEASE.md for exact commands. The final local run is outputs/m5_runs/m6_final_20260616, and the dashboard command is .venv/bin/python -m streamlit run dashboard/streamlit_app.py --server.port 8504.